

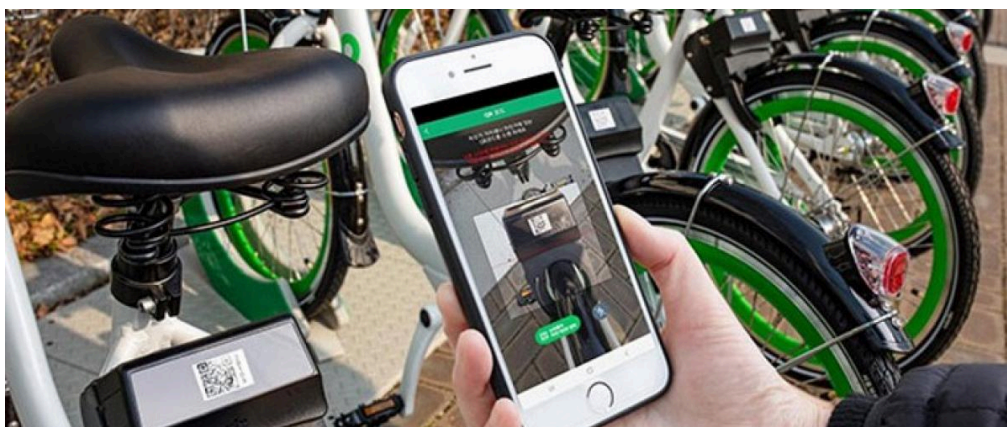
Visit also

ADB Knowledge Events

Visit also **ADB Knowledge
Events**

CASE STUDY

How Seoul Eased Traffic Congestion and Reduced Pollution Through Bike Sharing



On-board devices and QR-based payment systems are installed on all bicycles to improve user convenience and increase bicycle usage. Photo credit: Seoul Urban Solutions Agency.

Share on:



Published: 30 May 2023

The government implemented a communication plan, expanded bike roads, and improved user convenience to increase bicycle usage as a public transport alternative.

Overview

Ttareungyi Bike, Seoul's public share bike network, was officially launched in 2015, after a year of pilot

operations. Since then, it has had enormous impact in terms of relieving traffic congestion and mitigating emissions. It currently has a fleet of 40,500 bicycles and 2,600 rental stations across the city.

The Ttareungyi system has two unique characteristics:

- It is a public share bike system. The bicycles are owned by the city and used by its citizens.
- The users can use the bicycles to get to their varied destinations, unlike the general public transportation that has a fixed point of destination.

The Ttareungyi system delivers benefits of both public transportation and personal mobility for citizens. The Seoul Metropolitan Government established and implemented a mid- to long-term comprehensive plan for bicycle use proliferation, and established a dedicated department within the local government to operationalize and manage the city's public bikes. The Bicycle Policy Division is composed of four teams: the Bicycle Policy Team, the Public Bicycle Team, Bicycle Road Team, and the Bicycle Facilities Team.

Challenges

Seoul was faced with problems of severe traffic congestion during morning and evening peak times, and high levels of greenhouse gas (GHG) emissions from the transport sector. The city had reached a point where traffic congestion could no longer be addressed by adding new roads. It was necessary to find innovative approaches to increase the use of public transit.

Context

Seoul emits 4.6 million tons of greenhouse gases per year, with 19.2% of total emissions stemming from the transport sector. Given the severity of the city's climate emissions profile, transitioning toward a more sustainable urban mobility network is high priority for the Seoul Metropolitan Government. This is emphasized in the city's Comprehensive Plan for Climate Action announced in 2022, which articulates its commitment toward becoming a carbon neutral city by 2050 through strategic policy priorities in buildings, mobility, green spaces, urban safety and resiliency, and citizen cooperation.

Public share bikes provide a climate-aligned mechanism to ensure that mobility rights are

guaranteed for all citizens, regardless of where they are located, the time of day, or their destination.

Solutions

The Seoul Metropolitan Government started the Ttareungyi Bike to promote bicycle usage to resolve a host of interconnected urban challenges— environmental pollution and GHG emissions, high energy consumption, and traffic congestion. Due to the complexity of the challenges, the city focused both on expanding the supply of public share bikes and bicycle infrastructure, and implementing a variety of supporting activities and initiatives.

From the policy perspective, it implemented the following:

- It established a comprehensive mid- to long-term bicycle usage promotion plan to communicate the strategic framework that would govern bicycle promotion in Seoul, and to institutionalize public share bike operations and management as a policy priority. It instituted a governance framework to work directly with citizens and community groups to gather opinions and communicate bicycle usage/safety campaigns.
- It instituted a regulatory reform package to proactively address issues that would emerge as bicycle usage became more prevalent, including bicycle-only lane regulations, bicycle speed limits, bicycle insurance policies, among others.
- It integrated the Ttareungyi bicycle system’s operational data and information into its Transport Analysis and Index Management System (TAIMS)—a data-driven transport sector policy decision-making support system—in order to manage Ttareungyi operations and planning more accurately and efficiently, including decisions on where to install bicycle rental stations, bicycle lanes, and operational monitoring.

From an operational standpoint, the Seoul Metropolitan Government focused on improving user convenience to increase bicycle usage. For instance, on-board devices and QR-based payment systems were installed on all bicycles. A dedicated Ttareungyi mobile app was developed to facilitate more convenient reservation and search system for bicycle rental station locations.

In terms of bicycle road infrastructure, the Seoul Metropolitan Government created the 2026 Mid-Long Term Bicycle Road Network Construction Plan.

It laid out plans to connect all disjointed sections of major bicycle roads throughout the city, and ensure that all main areas of the city are accessible via bicycle. The plan also ensures that a sufficient level of budget is allocated towards bicycle road safety facility expansions.



One of the bicycle safety facilities the city government implemented was the bicycle lane differentiation using LEDs. Photo credit: Seoul Urban Solutions Agency.

Results

Since its introduction in 2015, the Ttareungyi system has made enormous quantifiable strides in promoting bicycle usage as an alternative urban transit mode:

- Users of the Ttareungyi’s dedicated app increased to 668,725 in 2019 from 34,162 in 2015.
- Bicycle rental rose to 13.7 million in 2021 from 3.6 million in 2016, which translates to over 100 million individual rides and 275.3 million kilometers of distance traveled.

The system also helped offset about 2,000 tons of carbon emissions with more people choosing to use bikes instead of motor vehicles.

Building on the momentum of the Ttareungyi system, the Seoul Metropolitan Government plans to include it into the wider public transport network via a public bike–public transit integrated transfer platform to enable the implementation of an improved public transit transfer mileage and fare system. This would ultimately enhance the convenience and usage rates of the Ttareungyi, as well as other public transit options.

The Seoul Metropolitan Government also plans to expand urban bicycle lanes to 194.42 kilometers and strengthen the bicycle road network by 2026.

Lessons

One of the weaknesses of bicycle usage is its heavy dependence on good weather. People are generally averse to traveling via bicycle in heavy rain, wind, snow, and heat. Thus, it is difficult for public bicycles to completely replace public transit or vehicles as a dominant mode of transit on a city-wide scale. The Seoul Metropolitan Government implemented various initiatives to overcome the weather concern, including investing in bicycle-related infrastructure and facilities.

The program also faced operational challenges, such as bicycle theft, poor bicycle return rates, and inconvenient reservation/payment systems. A survey during the pilot stage of the Ttareungyi system indicated complaints about difficulties in using the dedicated app, payment system not interlinked with the city transportation card, lack of rental stations around subway or bus stations, and poor layout or insufficient bicycle road infrastructure.

In response, the Seoul Metropolitan Government rolled out two system upgrades: QR code-based payment system, and anti-theft devices. Aside from the system upgrades, it also added bicycle storage facilities, increased bicycle fleet, and expanded bicycle roads.

Through the improvements, various challenges and inconveniences of the Ttareungyi system are being addressed in stages, and usage of the bikes has been increasing rapidly since their introduction in 2015.

It should also be noted that the citizen survey during the pilot stage showed a high rate of support for the bike system as a whole, with over 76% of surveyed citizens expressing high levels of satisfaction with the system, and 99% indicating that they would prefer for the system to be expanded.

Resources

B. Eun-ji. 2021. [Seoul City's Bike-Sharing Service Becomes Safer with Soaring Demand](#). *Korea Times*. 31 March.

S. Khatib. 2022. [How to Use Seoul's Public Bikes, Ttareungyi](#). *Korea by Me*. 11 February.

Seoul Metropolitan Government. [Seoul Public Bike, Ttareungyi](#).

Ssunha. 2017. [Seoul Public Bike, Ttareungyi Becomes More Convenient to Use](#). *Seoul Solution*. 11 July.



Sujin Kim

Senior Transport Specialist, Seoul Urban Solutions Agency

Sujin Kim leads in the design and delivery of urban transport policy and infrastructure advisory services for partner cities across the globe. He has over 20 years of experience in the design and delivery of transport policy planning, transport infrastructure development and operations, traffic safety management, and ITS design and implementation.

Follow Sujin Kim on



Seoul Urban Solutions Agency

The **Seoul Urban Solutions Agency** (SUSA) was established by the Seoul Metropolitan Government to share its urban development experiences with other cities seeking to become sustainable and smart urban domains. Through a wide network of partnerships with the public and private sectors within and outside the Republic of Korea, SUSA works to connect and leverage its wide range of knowledge and resources to assist in solving the development challenges of its partner cities.

Leave your question or comment in the section below:

YOU MIGHT ALSO LIKE

Car-Free Days Can Give Big Cities Room to Breathe

Case Study

A Strategy for Creating a Bicycle-Friendly City

Case Study

Creating Green Spaces in a Congested City

Case Study

📍 Korea, Republic of; Climate change; Transport; Urban development

🏠 SDG: Goal 3: Good Health and Well-being; Goal 11: Sustainable Cities and Communities; Goal 13: Climate Action

Disclaimer

The views expressed on this website are those of the authors and do not necessarily reflect the views and policies of the Asian Development Bank (ADB) or its Board of Governors or the governments they represent. ADB does not guarantee the accuracy of the data included in this publication and accepts no responsibility for any consequence of their use. By making any designation of or reference to a particular territory or geographic area, or by using the term “country” in this document, ADB does not intend to make any judgments as to the legal or other status of any territory or area.



Subscribe to Development Asia's
Newsletter.

Follow us

